ME602T - Assignment 2

When the program is run an option to choose whether to use matrix multiplication, finding determinant or to exit the program will be given.
Choosing `1` will enter matrix multiplication function.
Choosing `2` will enter determinant and transpose function.
Choosing `-1` will exit from the program.

Matrix Multiplication

The matrix multiplication function prompts user to enter the order of two matrices. If the multiplication cannot be performed, `Cannot multiply the matrices with given order.` will be printed and the program will exit.
If multiplication is possible, the user is then prompted to enter the matrix elements row wise. The matrices are stored column-wise to facilitate easier column access required for the multiplication algorithm.
Consider the multiplication `C = AB`, where `A` is an $l \times m$ matrix and `B` is a $m \times n$ matrix. we know each column of `C` is a linear combination of columns of `A` and is written as

$$c_j = Ab_j = \sum_{k=1}^{m} b_{kj} a_k$$

These operations are done using `Add` and `ScalarMult` helper functions.

Example test cases and their outputs:
Case 1: 2x2 square matrices

```
Select matrix multiplication or determinant of matrix.
Enter 1 for matrix multiplication or 2 for determinant of matrix or -1 to
exit: 1
Enter the order of matrix A: 2 2
Enter the order of matrix B: 2 2
Enter elements of matrix A row wise:
1 2
3 4
Enter elements of matrix B row wise:
5 6
7 8
Matrix multiplication result of given matrices is :
19  22
43  50
```

Case 2: 2x3 and 3x2 matrices

```
Select matrix multiplication or determinant of matrix.
Enter 1 for matrix multiplication or 2 for determinant of matrix or -1 to
exit: 1
Enter the order of matrix A: 2 3
Enter the order of matrix B: 3 2
Enter elements of matrix A row wise:
1 2 3
4 5 6
Enter elements of matrix B row wise:
7 8
9 10
11 12
Matrix multiplication result of given matrices is :
58  64
139  154
```

Case 3: Incompatible dimensions

```
Select matrix multiplication or determinant of matrix.
Enter 1 for matrix multiplication or 2 for determinant of matrix or -1 to
exit: 1
Enter the order of matrix A: 2 2
Enter the order of matrix B: 3 2
Cannot multiply the matrices with given order.
```

Determinant and Transpose of Square Matrix

The determinant is found using gaussian elimination method which turns the given matrix into an upper triangular matrix. In an upper triangular matrix all elements below the principal diagonal elements are zero. This is achieved by using row operations. If a principal diagonal element is zero, then that row is swapped with a row below it, whose principal diagonal element is non-zero. When a row swap happens, the variable swaps is incremented. The final determinant of the matrix is then multiplied with $(-1)^{swaps}$.

The transpose of matrix is found by swapping the ij^{th} element with ji^{th} element. The transpose is created and stored as a separate matrix.

Example test cases and their outputs:
Case 1: 2x2 matrix

```
Select matrix multiplication or determinant of matrix.
Enter 1 for matrix multiplication or 2 for determinant of matrix or -1 to
exit: 2
Enter order of matrix: 2
Enter the matrix elements row wise:
3 8
4 6
```

```
Determinant of given matrix is -14
Transpose of the given matrix is:
3    4
8    6
```

Case 2: 3x3 matrix

```
Select matrix multiplication or determinant of matrix.
Enter 1 for matrix multiplication or 2 for determinant of matrix or -1 to
exit: 2
Enter order of matrix: 3
Enter the matrix elements row wise:
1 2 3
0 1 4
5 6 0
Determinant of given matrix is 1
Transpose of the given matrix is:
1    0    5
2    1    6
3    4    0
```

Case 3: 1x1 matrix

```
Select matrix multiplication or determinant of matrix.
Enter 1 for matrix multiplication or 2 for determinant of matrix or -1 to
exit: 2
Enter order of matrix: 1
Enter the matrix elements row wise:
5
Determinant of given matrix is 5
Transpose of the given matrix is:
5
```